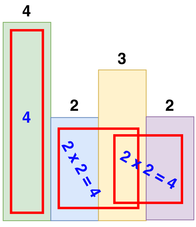
Largest Rectangle in Histogram

Problem Statement

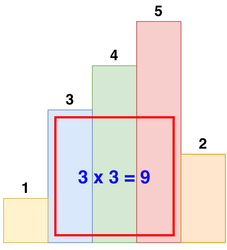
Given an array of integers `heights`, where `heights[i]` represents the height of the histogram's bar, return the area of the `largest rectangle` in the histogram. Each bar has width equal to `1`.

Examples

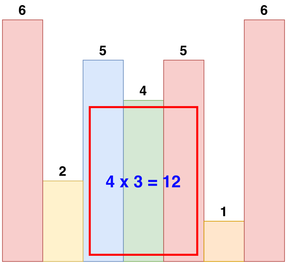
- **Example 1**:
  - **Input**: `[4, 2,0,3,2]`



- **Expected Output**: `4`
- **Justification**: The largest rectangle can be formed by using the bar with height `4` only or `4` and `2` or `3` and `2`, covering a total area of `2x2=4`.
- **Example 2**:
  - **Input**: `[1,3,4,5,2]`



- **Expected Output**: `9`
- **Justification**: The largest rectangle can be drawn from the second bar to the fourth bar, having height `3`, resulting in an area of `3x3=9`.
- **Example 3**:
  - **Input**: `[6,2,5,4,5,1,6]`



- **Expected Output**: `12`
- **Justification**: The maximum rectangular area is formed by the bars of heights `5`, `4`, `5`, which is equal to `4x3=12`. This rectangle spans from the third to the sixth bar.

Try it yourself

Try solving this question here:

Java

```java
import java.util.Stack;

public class Solution {
    // Method to calculate the largest rectangle area
    public int largestRectangleArea(int[] heights) {
        // ToDo: Write Your Code Here.
        return 0; // Return the maximum area
    }
}
```

Test Case | Run Code Result | Stdout

[4,2,0,3,2]

Use Example Testcases | Run | Submit